

Solutions to MP363 Problem set 5

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P.1

We know from the postulates of QM that $|\psi(t, x)|^2$ is interpreted as the probability density of a QM particle, namely, the probability of finding a particle described by this wavefunction in a region R at time t is $\int_R |\psi(t, x)|^2 dx$. This holds for any QM system in a stable potential well, so it certainly holds for a particle in the ground state of the QM SHO, i.e. one with wavefunction

$$\psi(t, x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar} - \frac{i\omega t}{2}}$$

Its probability density is then

$$|\psi(t, x)|^2 = \psi^*(t, x) \psi(t, x)$$

$$= \left[\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar} + \frac{i\omega t}{2}} \right] \left[\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar} - \frac{i\omega t}{2}} \right]$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}} e^{-\frac{m\omega x^2}{\hbar}}$$

Now, the classically forbidden region is given by $x < -\sqrt{\frac{2E_0}{m\omega}} = -\sqrt{\frac{\hbar}{m\omega}}$ and $x > \sqrt{\frac{2E_0}{m\omega}} = \sqrt{\frac{\hbar}{m\omega}}$, so the probability of finding the particle in this region is

$$p(\text{classically forbidden}) = \int_{|x| > \sqrt{\frac{\hbar}{m\omega}}} |\psi(t, x)|^2 dx$$

$$= \int_{-\infty}^{-\sqrt{\frac{\hbar}{m\omega}}} \sqrt{\frac{m\omega}{\pi\hbar}} e^{-\frac{m\omega x^2}{\hbar}} dx + \int_{\sqrt{\frac{\hbar}{m\omega}}}^{\infty} \sqrt{\frac{m\omega}{\pi\hbar}} e^{-\frac{m\omega x^2}{\hbar}} dx.$$

To make this a bit more simple-looking, let $s = \sqrt{\frac{m\omega}{\hbar}} x$ be a new integration variable. This gives

$$p(\text{classically forbidden}) = \int_{-\infty}^{-1} \sqrt{\frac{m\omega}{\pi\hbar}} e^{-s^2} \left(\sqrt{\frac{\hbar}{m\omega}} ds\right) + \int_1^{\infty} \sqrt{\frac{m\omega}{\pi\hbar}} e^{-s^2} \left(\sqrt{\frac{\hbar}{m\omega}} ds\right)$$

$$= \frac{1}{\sqrt{\pi}} \left(\int_{-\infty}^{-1} e^{-s^2} ds + \int_1^{\infty} e^{-s^2} ds \right)$$

$$= \frac{2}{\sqrt{\pi}} \int_1^{\infty} e^{-s^2} ds$$

because e^{-s^2} is an even function, so $\int_{-\infty}^{-1} e^{-s^2} ds = \int_1^{\infty} e^{-s^2} ds$.

This can be related to the complementary error function $\text{erfc}(x)$, which

(2)

is defined as

$$\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-s^2} ds$$

so that we see that

$$p(\text{classically forbidden}) = \operatorname{erfc}(1)$$

You can use whatever source you like to find this value, or numerically compute it yourself. I used the Wolfram functions website, specifically,

<http://functions.wolfram.com/GammaBetaErf/Erfc/FunctionEvaluation.jsp?name=Erfc>.

By putting $z=1$ into $\operatorname{erfc}(z)$ and clicking the "evaluate" button, I get (to 500 digits of precision)

$$\operatorname{erfc}(1) \approx 0.157299$$

so that there is a 15.73% chance that the QM SHO groundstate particle will be in the classically forbidden region. This is actually quite high - nearly 1 out of every 6 particles will be found in this region! (Yet another way that QM gives results contrary to classical mechanics...)

P. 2

$\langle \psi | a | \psi \rangle = \langle a^\dagger \psi | \psi \rangle$ for any system, but $a^\dagger$ is indeed the adjoint of a , then $\langle \psi | a | \psi \rangle = \langle a^\dagger \psi | \psi \rangle$ should be so. Let's compute this difference and show that this is indeed the case:

$$\begin{aligned} \langle \psi | a | \psi \rangle - \langle a^\dagger \psi | \psi \rangle &= \int_{-\infty}^{\infty} \psi^*(x) [a\psi(x)] dx - \int_{-\infty}^{\infty} [a^\dagger \psi(x)]^* \psi(x) dx \\ &= \int_{-\infty}^{\infty} \psi^*(x) \left[\sqrt{\frac{m\omega}{2\hbar}} (x\psi(x) + \frac{\hbar}{m\omega} \frac{\partial \psi}{\partial x}(x)) \right] dx - \int_{-\infty}^{\infty} \left[\sqrt{\frac{m\omega}{2\hbar}} (x\psi(x) - \frac{\hbar}{m\omega} \frac{\partial \psi}{\partial x}(x)) \right]^* \psi(x) dx \\ &= \sqrt{\frac{m\omega}{2\hbar}} \left(\int_{-\infty}^{\infty} \psi^*(x) x \psi(x) dx + \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} \psi^*(x) \frac{\partial \psi}{\partial x}(x) dx \right. \\ &\quad \left. - \int_{-\infty}^{\infty} x \psi^*(x) \psi(x) dx - \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} \frac{\partial \psi^*}{\partial x}(x) \psi(x) dx \right). \end{aligned}$$

The first and third terms add to zero, so we are left with

$$\langle \psi | a | \psi \rangle - \langle a^\dagger \psi | \psi \rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\int_{-\infty}^{\infty} \psi^* \frac{\partial \psi}{\partial x} dx + \int_{-\infty}^{\infty} \frac{\partial \psi^*}{\partial x} \psi dx \right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{\infty} \left(\psi^* \frac{\partial \psi}{\partial x} + \frac{\partial \psi^*}{\partial x} \psi \right) dx$$

$$\begin{aligned}
 & \textcircled{3} \\
 & = \sqrt{\frac{\hbar}{2m\omega}} \int_{-\infty}^{\infty} \frac{\partial}{\partial x} (\psi^*(x) \psi(x)) dx \\
 & = \sqrt{\frac{\hbar}{2m\omega}} \left[\psi^*(x) \psi(x) \right]_{-\infty}^{\infty}
 \end{aligned}$$

And now we use the fact that we've used time and time again: if ψ & ψ^* are normalizable states, then both must go to zero as $x \rightarrow \pm\infty$. Thus, the quantity above vanishes, giving $\langle \psi | a | \psi \rangle = \langle a^\dagger \psi | \psi \rangle$. Thus, therefore, proves that $a^\dagger$ is the adjoint of a .

P.3

Let's get the hint out of the way first: X and P are related to a and $a^\dagger$ via

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(X + \frac{i}{m\omega} P \right), \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(X - \frac{i}{m\omega} P \right)$$

We can easily see from here that

$$X = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger), \quad P = -i\sqrt{\frac{m\hbar\omega}{2}} (a - a^\dagger).$$

This, together with

$$a|n\rangle = \sqrt{n+1}|n+1\rangle, \quad a^\dagger|n\rangle = \sqrt{n}|n-1\rangle$$

given

$$X|n\rangle = \sqrt{\frac{\hbar}{2m\omega}} (a|n\rangle + a^\dagger|n\rangle) = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n}|n-1\rangle + \sqrt{n+1}|n+1\rangle)$$

and

$$P|n\rangle = -i\sqrt{\frac{m\hbar\omega}{2}} (a|n\rangle - a^\dagger|n\rangle) = -i\sqrt{\frac{m\hbar\omega}{2}} (\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle).$$

Now, for the expectation values, we have

$$\langle X \rangle = \langle n | X | n \rangle \quad \text{and} \quad \langle P \rangle = \langle n | P | n \rangle$$

and so if we remember that $\langle n | n' \rangle = \delta_{nn'}$ (i.e. the states are orthonormalized), then

$$\begin{aligned}
 \langle X \rangle &= \langle n | X | n \rangle = \left\langle n \left| \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n}|n-1\rangle + \sqrt{n+1}|n+1\rangle) \right| \right\rangle \\
 &= \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n} \langle n | n-1 \rangle + \sqrt{n+1} \langle n | n+1 \rangle) \\
 &= \textcircled{0}
 \end{aligned}$$

$$\begin{aligned}
 & \text{here } \langle n | n-1 \rangle = \langle n | n+1 \rangle = 0. \text{ Similarly,} \\
 \langle P \rangle &= \langle n | P | n \rangle = \left\langle n \left| -i\sqrt{\frac{m\hbar\omega}{2}} (\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle) \right| \right\rangle
 \end{aligned}$$

(c)

$$= -i\sqrt{\frac{\hbar m \omega}{2}} (\sqrt{n} \langle n|n-1 \rangle - \sqrt{n+1} \langle n|n+1 \rangle)$$

$$= 0$$

for the same reason. To find $\langle X^2 \rangle$ and $\langle P^2 \rangle$, we need to see what $X^2|n\rangle$ and $P^2|n\rangle$ are; we already have $X|n\rangle$, so

$$X^2|n\rangle = X(X|n\rangle) = X\left(\sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n}|n-1\rangle + \sqrt{n+1}|n+1\rangle)\right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}} X|n-1\rangle + \sqrt{\frac{(n+1)\hbar}{2m\omega}} X|n+1\rangle$$

But $X|n-1\rangle$ is just $X|n\rangle$ with n replaced by $n-1$, i.e.

$$X|n-1\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n-1}|n-2\rangle + \sqrt{n}|n\rangle)$$

and similarly $X|n+1\rangle$ is obtained by replacing n by $n+1$:

$$X|n+1\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n+1}|n\rangle + \sqrt{n+2}|n+2\rangle)$$

so that

$$X^2|n\rangle = \sqrt{\frac{\hbar}{2m\omega}} \left[\sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n-1}|n-2\rangle + \sqrt{n}|n\rangle) + \sqrt{\frac{(n+1)\hbar}{2m\omega}}(\sqrt{n+1}|n\rangle + \sqrt{n+2}|n+2\rangle) \right]$$

$$= \frac{\hbar}{2m\omega} \left(\sqrt{n(n-1)}|n-2\rangle + (2n+1)|n\rangle + \sqrt{(n+1)(n+2)}|n+2\rangle \right)$$

Since $\langle n|n-2\rangle = \langle n|n+2\rangle = 0$ but $\langle n|n\rangle = 1$, we therefore get

$$\langle n|X^2|n\rangle = \langle X^2 \rangle$$

$$= \frac{\hbar}{2m\omega} (\sqrt{n(n-1)} \cdot 0 + (2n+1) \cdot 1 + \sqrt{(n+1)(n+2)} \cdot 0)$$

$$= \boxed{\frac{\hbar}{m\omega} \left(n + \frac{1}{2}\right)}$$

and thus

$$\Delta x = \sqrt{\langle X^2 \rangle - \langle X \rangle^2}$$

$$= \boxed{\sqrt{\frac{\hbar}{m\omega} \left(n + \frac{1}{2}\right)}}$$

we find $P^2|n\rangle$ in a similar fashion:

$$P^2|n\rangle = P(P|n\rangle)$$

$$= P\left[-i\sqrt{\frac{\hbar m \omega}{2}}(\sqrt{n}|n-1\rangle - \sqrt{n+1}|n+1\rangle)\right]$$

$$\begin{aligned}
 &= -i\sqrt{\frac{n m \hbar \omega}{2}} P |n-1\rangle + i\sqrt{\frac{(n+1) m \hbar \omega}{2}} P |n+1\rangle \\
 &= -i\sqrt{\frac{n m \hbar \omega}{2}} \left[-i\sqrt{\frac{\hbar \omega}{2}} (\sqrt{n-1} |n-2\rangle - \sqrt{n} |n\rangle) \right] \\
 &\quad + i\sqrt{\frac{(n+1) m \hbar \omega}{2}} \left[-i\sqrt{\frac{\hbar \omega}{2}} (\sqrt{n} |n\rangle - \sqrt{n+2} |n+2\rangle) \right] \\
 &= \frac{m \hbar \omega}{2} \left[-\sqrt{n(n-1)} |n-2\rangle + (2n+1) |n\rangle - \sqrt{(n+1)(n+2)} |n+2\rangle \right]
 \end{aligned}$$

so

$$\langle P^2 \rangle = \langle n | P | n \rangle$$

$$= \frac{m \hbar \omega}{2} \left[-\sqrt{n(n-1)} \cdot 0 + (2n+1) \cdot 1 - \sqrt{(n+1)(n+2)} \cdot 0 \right]$$

$$= m \hbar \omega \left(n + \frac{1}{2} \right)$$

and thus

$$\Delta p = \sqrt{\langle P^2 \rangle - \langle P \rangle^2} = \boxed{\sqrt{m \hbar \omega \left(n + \frac{1}{2} \right)}}$$

So both Δx and Δp increase as n increases: the higher-energy states have larger uncertainties in both position and momentum.

Interestingly, note that

$$\Delta x \Delta p = \hbar \left(n + \frac{1}{2} \right),$$

so the ground state - which has $n=0$ - obeys $\Delta x \Delta p = \frac{\hbar}{2}$, the absolute minimum value $\Delta x \Delta p$ can have. Thus, the ground-state of the QM SHO is called a "minimum uncertainty state".